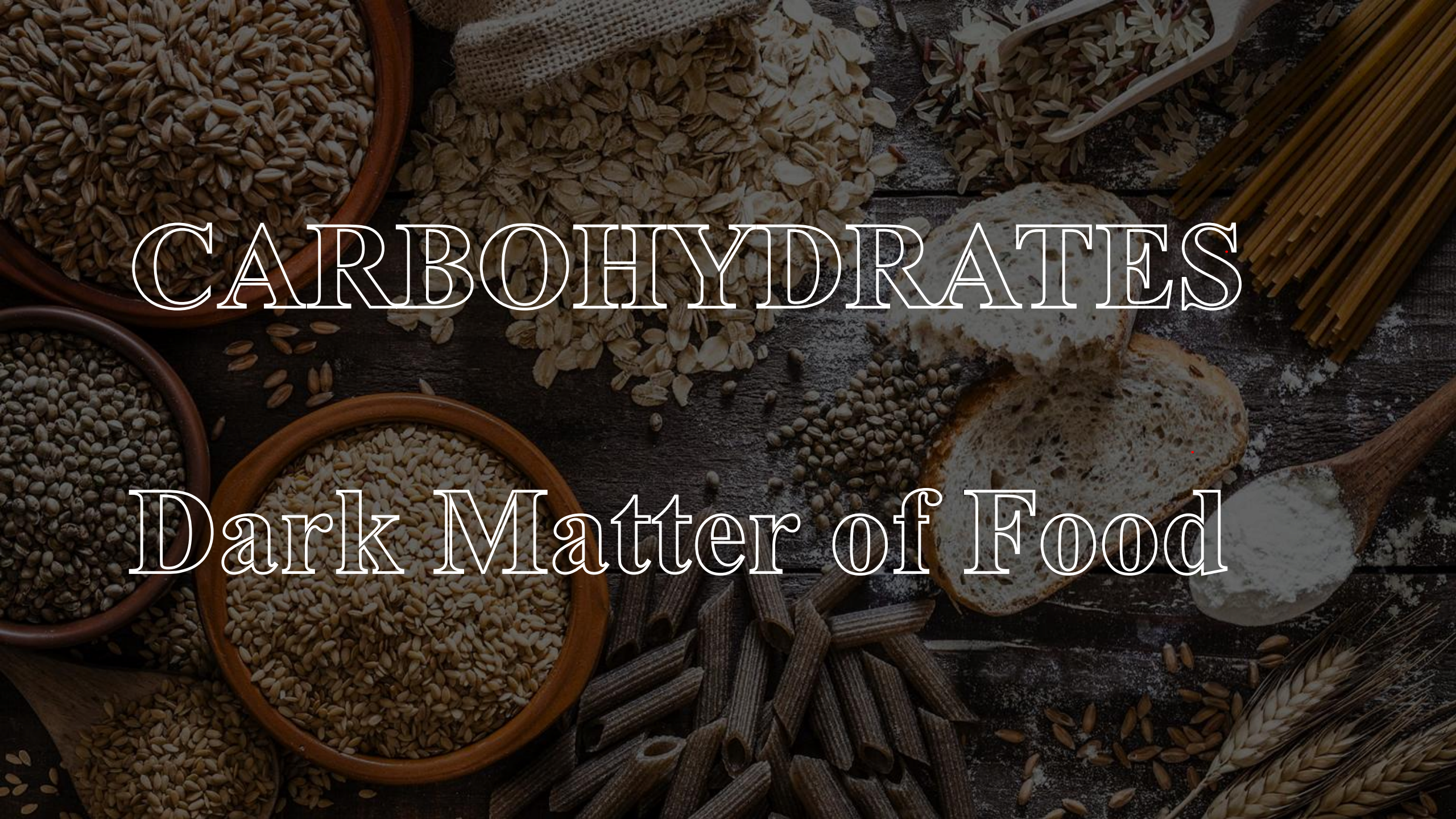


A top-down view of various grains and carbohydrates arranged on a dark, textured wooden surface. In the top left, a wooden bowl is filled with whole wheat berries. Next to it is a pile of rolled oats, some of which are spilling out. To the right, there's a small pile of white rice grains and a bundle of uncooked spaghetti. In the bottom left, another wooden bowl contains small, light-colored grains, possibly quinoa or millet. In the bottom center, a bowl is filled with whole grain barley. To the right of this, there's a pile of uncooked penne pasta. Further right, two slices of rustic, whole-grain bread are visible. A wooden spoon filled with white flour sits in the bottom right corner. Scattered throughout the scene are various other grains, including wheat berries, and some wheat stalks in the bottom right corner.

CARBOHYDRATES

Dark Matter of Food

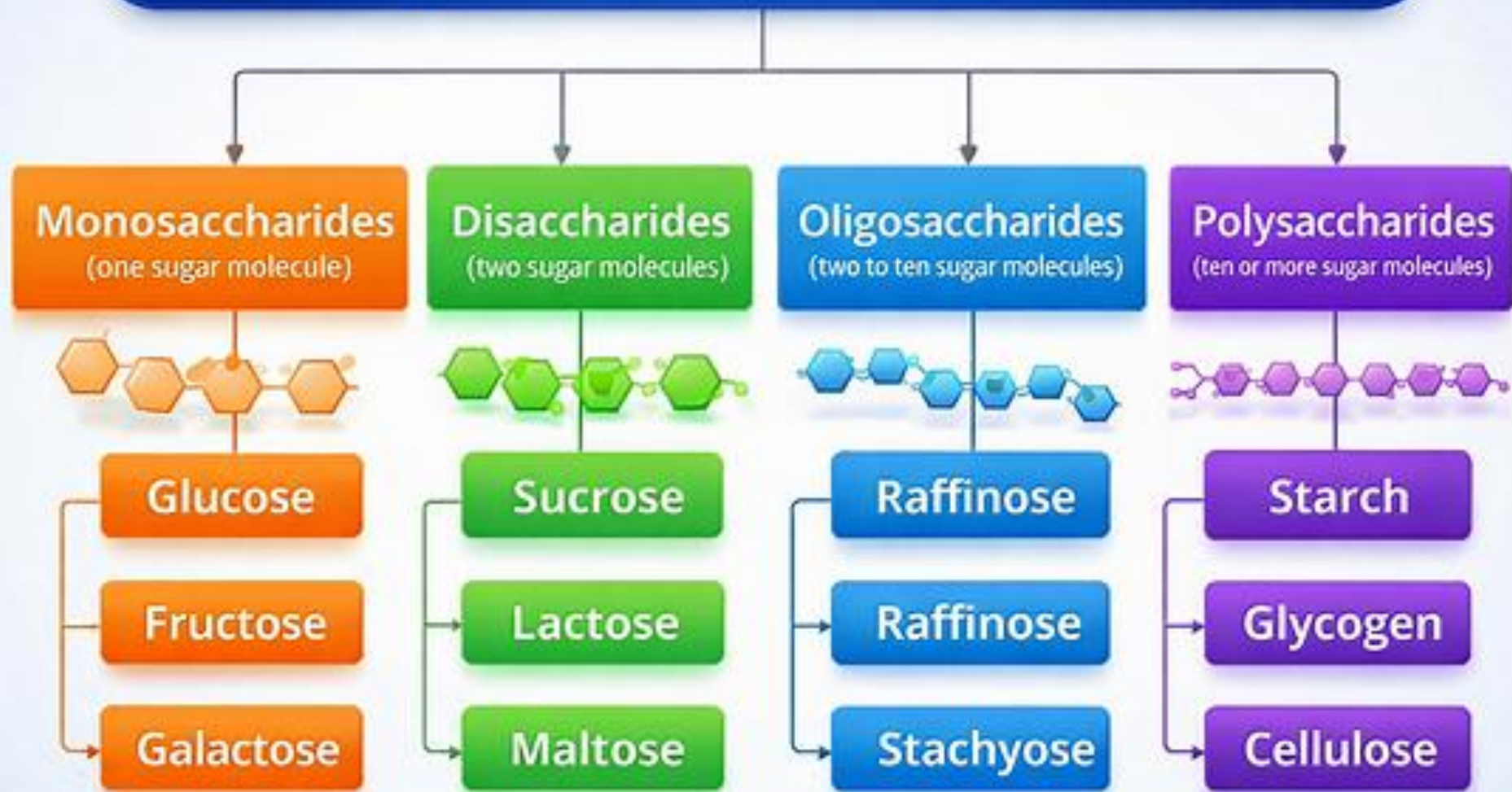


CHEMISTRY ?

Carbohydrates

Hydrates of Carbon

Carbohydrate Classification Chart



Classification and Sources of Carbohydrates

Free Sugars – Monosaccharides and Disaccharides

Monosaccharides

- **Simplest form of carbohydrates** and these are the monomeric units of large carbohydrates
- **Glucose** (called as dextrose / grape sugar / corn sugar) is found in sweet fruits such as **grapes, berries and oranges**; vegetables like sweet corn and carrot
- Glucose is the form of carbohydrate circulating in the blood and is utilised by the cells for energy
- **Fructose** (fruit sugar) is found in **honey, ripe fruits and some vegetables**; basic component of table sugar; used as sweetener in processed foods and beverages – has a low impact on the blood sugar levels – Jack fruit, apple, grapes, pears and raisins are rich sources
- Fructose is metabolised by the liver and converted into energy sources without the intervention of insulin.
This is also converted into **fat**

Classification and Sources of Carbohydrates

Disaccharides

- **Two molecules of monosaccharides**, which can be digested further (or by acid hydrolysis)
- **Sucrose** (cane sugar: glucose + fructose)
- **Lactose** (milk sugar: glucose + galactose)
- **Maltose** (malt sugar: glucose + glucose)

- **Sugar alcohols (polyols):** These are not digested or absorbed in the small intestine but fermented by large gut microflora eg: *Erythritol*, *Maltitol*

Classification and Sources of Carbohydrates

Oligosaccharides/ Short Chain Carbohydrates (SCC)

- These are soluble in 80% ethanol
- Examples: **raffinose, stachyose, verbascose, inulin**
- These are not susceptible to hydrolysis by endogenous enzymes but fermented by microflora in the large intestine

Polysaccharides

- Not soluble in 80% ethanol under prescribed conditions
- Have more number of monomers linked by the glycosidic linkages at different positions
- The position of the glycosidic linkage determine the digestibility of these polysaccharides

Classification and Sources of Carbohydrates

There are different varieties of polysaccharides. They are:

Starch: This is of three types – **Rapidly digestible starch (RDS)**, **Slowly digestible starch (SDS)** and **resistant starch (RS)** – composition varies with food types and processing

- Starch is major storage polysaccharide of plants – found in cereal grains, potatoes and plantains
- This is major carbohydrate in human diet – contains *amylose and amylopectin*
- **RDS** is completely digested and absorbed in small intestine – this is an amorphous and dispersed form of starch – occurs in starchy foods that have been cooked by moist heat – **bread and potatoes**
- **SDS**– Completely digested but more slowly – accessibility to enzyme is the reason – partially milled grains and seeds (with dense structure) – granular starch in raw foods – **pasta**
- **Resistant starch** – starch which escapes digestion in small intestine and fermented in large intestine – prebiotic in nature and is a nutraceutical – found in legumes (*red gram, Bengal gram* – food processing alters the RS content)

Classification and Sources of Carbohydrates

Non-Starch Carbohydrates: (Majorly dietary fibre)

- Polysaccharides other than starch
- Also insoluble in 80% ethanol
- Plant cell walls are the major source – bran, stalks and leaves of vegetables
- *Hemicellulose, cellulose, lignins, pectins and gums*
- Some bacteria in the body can degrade these
- All those carbohydrates which are resistant to hydrolysis are collectively called as *dietary fibre*
- Resistant starch also functions as dietary fibre

$$\text{Total fibre} = \text{Dietary fibre} + \text{functional fibre}$$

Dietary fibre = collective terminology for complex indigestible carbohydrates

Functional fibre = isolated or extracted forms from natural sources / can be synthetic

Dietary Fibre Content

Food	Fiber (%weight)	Food	Fiber (%weight)
Almonds	3	Nuts	2
Wheat	3	Walnut	2
Lima beans	2	Broccoli	1
oatmeals	2	Carrot	1
Peach	2	Strawberry	1
Whole wheat flour	2	Apple	1
Corn	2	White flour	<1

**Current Daily Value (DV) for dietary fibre is 28 g
And is expressed in terms of percentage**

Top 10 Foods Highest in Fiber

28g of Fiber = 100% of the Daily Value (%DV)

1 Navy (Haricot) Beans



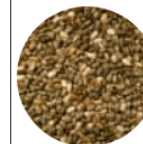
68% DV (19g)
per cup
255 calories

2 Avocados



48% DV (13g)
per avocado
322 calories

3 Chia Seeds



35% DV (10g)
per oz (~2 tbsps)
138 calories

4 Acorn Squash



32% DV (9g)
per cup cooked
115 calories

5 Green Peas



31% DV (9g)
per cup cooked
134 calories

6 Collard Greens



27% DV (8g)
per cup cooked
63 calories

7 Broccoli



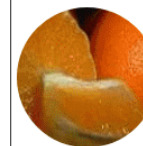
18% DV (5g)
per cup cooked
55 calories

8 Whole Wheat Pasta



16% DV (5g)
per cup
174 calories

9 Oranges



15% DV (4g)
per cup
85 calories

10 Sweet Potatoes



14% DV (4g)
per cup
114 calories

Glycemic Index and Glycemic Load

Glycemic Indices & Glycemic Loads of Common Foods		
Food	Glycemic Index	Glycemic Load*
White bread	72	10
Whole wheat bread	72	8
Baked potato	111	33
Sweet potato	61	11
Carrots	35	2
Honey	69	15
Watermelon	72	4
Banana	62	16
Grapes	59	11
Apple	39	6
Cranberry juice cocktail	68	24
Yogurt (low fat)	27	9
Skim milk	32	4
Whole milk	41	5

Glycemic index (GI):

- The glycemic index measures how quickly a carbohydrate-containing food raises blood glucose levels compared to pure glucose (which has a GI of 100)

Glycemic load:

- Glycemic load considers both the quality (GI) and quantity (amount of carbohydrate) in a serving of food.
- $$GL = \frac{GI \times \text{Carbohydrate content (g per serving)}}{100}$$

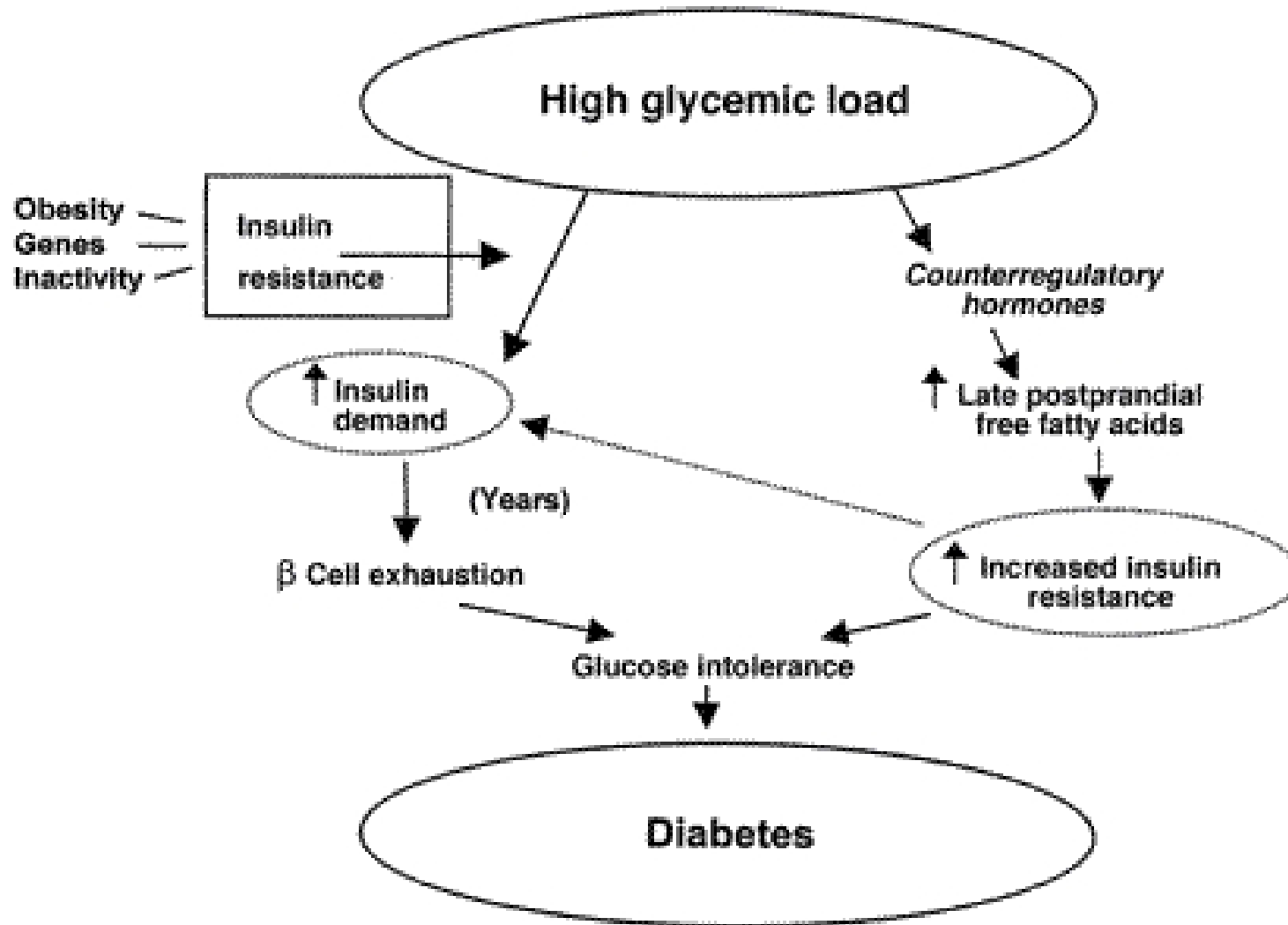
100

Low GL (≤ 10): Minimal impact on blood sugar.

Medium GL (11–19): Moderate impact.

High GL (≥ 20): Significant impact.

Glycemic Load and Diabetes



Better to consume fresh fruits rather than fruit juices

Glycemic load classification

- < 10 low GL
- 11 – 19 medium GL
- > 20 high

In case of diabetes, GL should be given importance rather than GI

Functions of Carbohydrates

Source of energy

- Least expensive source of energy
- Every gram of carbohydrate upon oxidation yields 4 kilo calories

Protein sparing action

- When carbohydrate is present in required quantity, proteins are used for tissue building and not for energy
- Dietary protein is used up for energy when carbohydrate intake is limited
- In such cases, tissue proteins are also broken down

Functions of Carbohydrates

Indispensable for nervous system

- Main source of energy for central nervous system is glucose
- Hypoglycaemia will result in brain damage

Role in muscle

- Carbohydrates are the major source of energy for muscle work; heart muscle also used glucose as energy source
- During muscle contraction, glycogen is broken down to lactic acid (glycolysis)
- During recovery period lactic acid is converted to acetyl CoA and then oxidised to CO_2 and H_2O , thus producing energy

Role in liver

- Detoxifying action and influence protein and fat metabolism
- Rate of oxidation of amino acids in liver is less if abundant carbohydrates are available

Functions of Carbohydrates

Synthesis of ribose from glucose

- Pentose for nucleic acid synthesis is obtained by metabolism of glucose (Hexose monophosphate pathway)
- Required for both DNA and RNA

Conversion to fat

- Excess carbohydrate is converted to fat and stored in adipose tissue
- When energy is required fat is released from adipose tissue

Promotes growth of desirable bacteria

- Lactose promotes growth of desirable bacteria
- Some microbes useful in the synthesis of vitamin B complex
- Lactose also enhances absorption of calcium

Functions of Carbohydrates

Contribution of dietary fibre

- Fibre stimulates peristaltic movement
- Prevents many degenerative diseases (Alzheimer's, diabetes, cardiovascular and others)

Structural component

- Carbohydrates are precursors for many organic compounds
- Glycoproteins and glycolipids as part of the cell wall structure
- Cellular functions such as cell growth adhesion and fertilisation
- Fibre of plants, exoskeleton of insects and cell wall of microorganisms